I.O.E

Module 2

DRIVERS BEHIND NEW NETWORK ARCHITECTURES

1.Building residential houses vs building a massive stadium..

2.The difference between IT and IoT networks is much like the difference between residential architecture and stadium architecture
3.The key difference between IT and IoT is the data.

4.IT systems are mostly concerned with reliable and continuous support of business applications such as email, web, databases, CRM systems, and so on.

5. IoT is all about the data generated by sensors and how that data is used.

6. The essence of IoT architectures thus involves how the data is transported, collected, analyzed, and ultimately acted upon

Challenge	Description	IoT Architectural Change Required
Scale	The massive scale of IoT endpoints (sensors) is far beyond that of typical IT networks.	The IPv4 address space has reached exhaustion and is unable to meet IoT's scalability requirements. Scale can be met only by using IPv6. IT networks continue to use IPv4 through features like Network Address Translation (NAT).
Security	IoT devices, especially those on wireless sensor networks (WSNs), are often physically exposed to the world.	Security is required at every level of the IoT network. Every IoT endpoint node on the network must be part of the overall security strategy and must support device-level authentication and link encryption. It must also be easy to deploy with some type of a zero-touch deployment model.
Devices and networks constrained by power, CPU, memory, and link speed	Due to the massive scale and longer distances, the networks are often constrained, lossy, and capable of supporting only minimal data rates (tens of bps to hundreds of Kbps).	New last-mile wireless technologies are needed to support constrained IoT devices over long distances. The network is also constrained, meaning modifications need to be made to traditional network-layer transport mechanisms.

The massive volume of data generated	The sensors generate a massive amount of data on a daily basis, causing network bottlenecks and slow analytics in the cloud.	Data analytics capabilities need to be distributed throughout the IoT network, from the edge to the cloud. In traditional IT networks, analytics and applications typically run only in the cloud.
Support for legacy devices	An IoT network often comprises a collection of modern, IP-capable endpoints as well as legacy, non-IP devices that rely on serial or proprietary protocols.	Digital transformation is a long process that may take many years, and IoT networks need to support protocol translation and/or tunneling mechanisms to support legacy protocols over standards-based protocols, such as Ethernet and IP.
The need for data to be analyzed in real time	Whereas traditional IT networks perform scheduled batch processing of data, IoT data needs to be analyzed and responded to in real-time.	Analytics software needs to be positioned closer to the edge and should support real-time streaming analytics. Traditional IT analytics software (such as relational databases or even Hadoop), are better suited to batch-level analytics that occur after the fact.

Security

- IT networks use firewall, IT endpoints are behind firewall
- IoT endpoints are often located in wireless sensor networks that use unlicensed spectrum and are not only visible to the world through a spectrum analyzer but often physically accessible
- IoT systems require consistent mechanisms of authentication, encryption, and intrusion prevention techniques

- IoT systems must:
- Be able to identify and authenticate all entities involved in the IoT service (that is, gateways, endpoint devices, home networks, roaming networks, service platforms)
- Ensure that all user data shared between the endpoint device and back-end applications is encrypted
- Comply with local data protection legislation so that all data is protected and stored correctly
- Take network-level approach to security in addition to device level approach

Constrained Devices and Networks

- Most IoT sensors are designed for a single job, and they are typically small and inexpensive
- They often have limited power, CPU, and memory, and they transmit only when there is something important
- The networks that provide connectivity also tend to be very lossy and support very low data rates.
- This is a completely different situation from IT networks
- IoT requires a new breed of connectivity technologies that meet both the scale and constraint limitations

Legacy Device Support

- Supporting legacy devices in an IT organization is not usually a big problem
- If someone's computer or operating system is outdated, she simply upgrades
- If someone is using a mobile device with an outdated Wi-Fi standard, such as 802.11b or 802.11g, you can simply deny him access to the wireless network, and he will be forced to upgrade

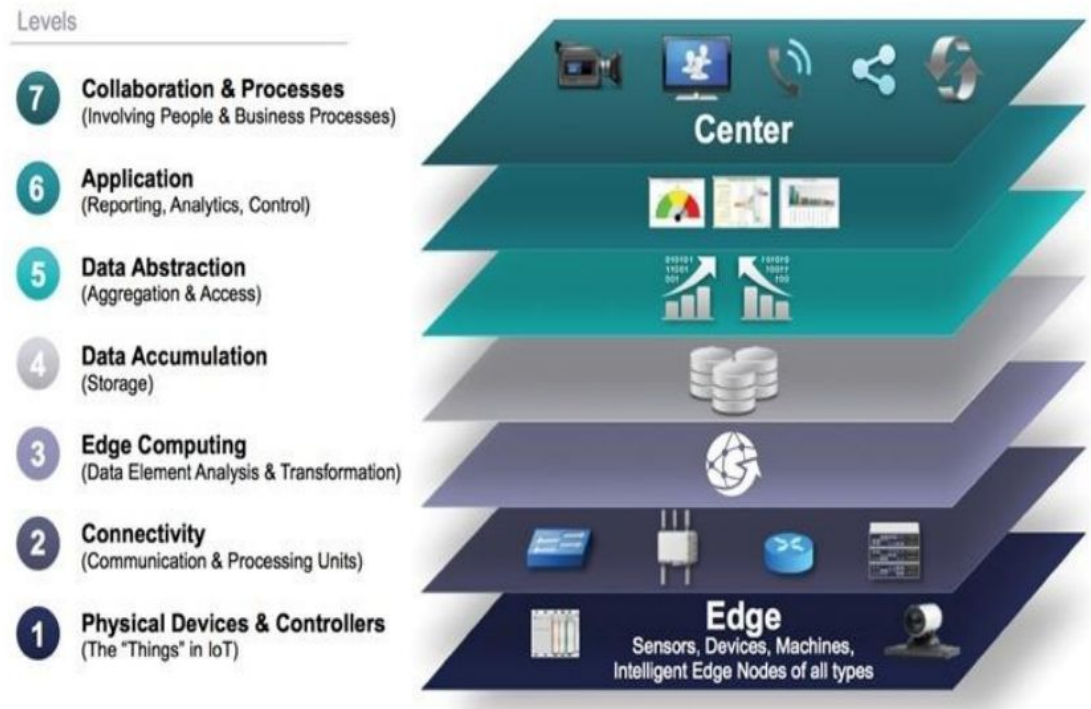
- In OT systems, end devices are likely to be on the network for a very long time—sometimes decades.
- As IoT networks are deployed, they need to support the older devices already present on the network, as well as devices with new capabilities

IOT WF Standardized Architecture

The IoT World Forum (IoTWF) Standardized Architecture

- A seven-layer IoT architectural reference model published by IoTWF architectural committee (Cisco, IBM, Rockwell Automation)
- Edge computing
- Data storage
- Access

IoT Reference Model Published by the IoT World Forum



- Using this reference model, we are able to achieve the following:
- Decompose the IoT problem into smaller parts
- Identify different technologies at each layer and how they relate to one another
- Define a system in which different parts can be provided by different vendors
- Have a process of defining interfaces that leads to interoperability
- Define a tiered security model that is enforced at the transition points between levels

Layer 1: Physical Devices and Controllers Layer

- The various endpoint devices and sensors that send and receive information
- The size of these "things" can range from almost microscopic sensors to giant machines in a factory.
- Their primary function is generating data and being capable of being queried and/or controlled over a network.

Layer 2: Connectivity Layer

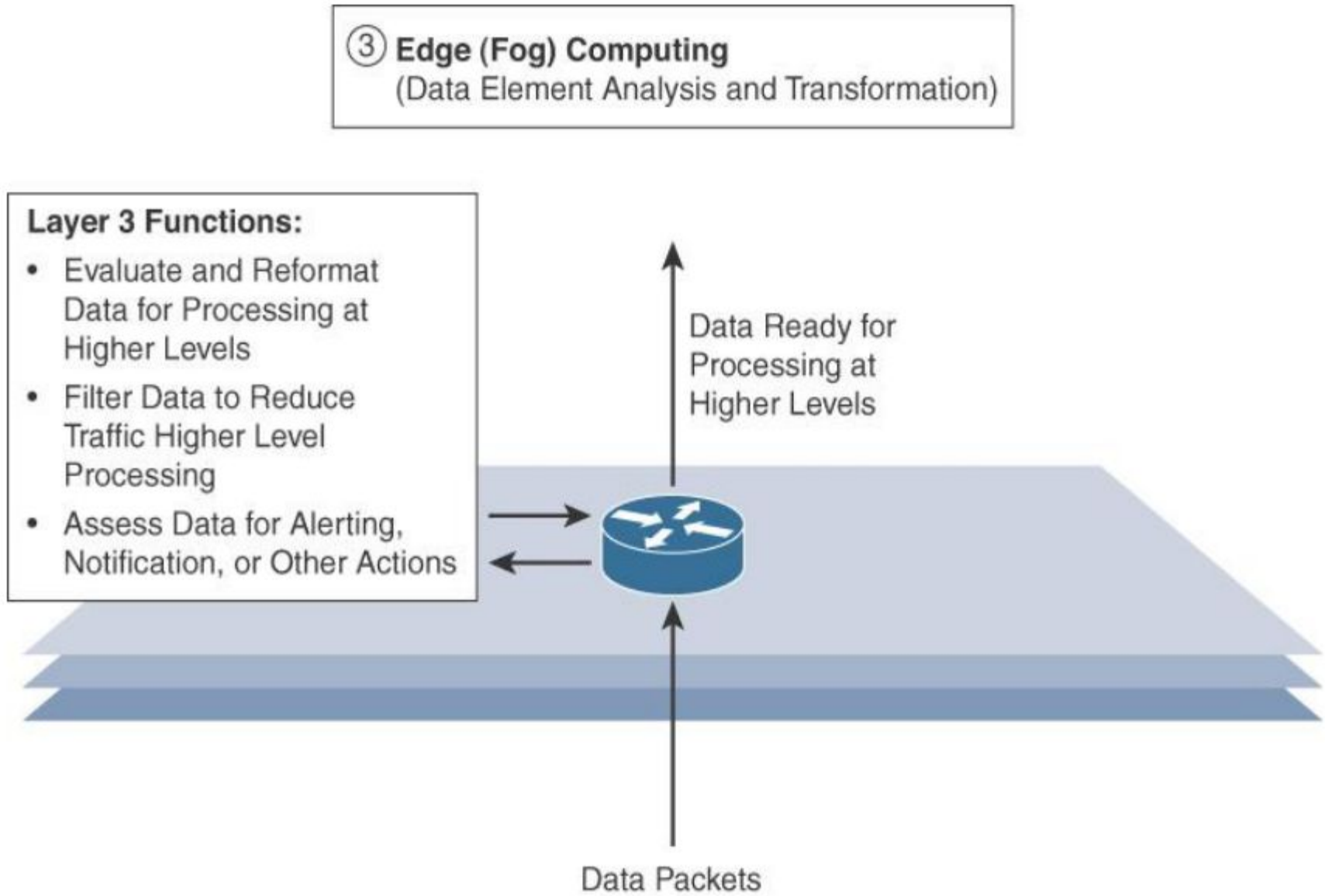
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Layer 3: Edge Computing Layer

- Fog layer
- At this layer, the emphasis is on data reduction and converting network data flows into information that is ready for storage and processing by higher layers
- One of the basic principles of this reference model is that information processing is initiated as early and as close to the edge of the network as possible
- Another important function that occurs at Layer 3 is the evaluation of data to see if it can be filtered or aggregated before being sent to a higher layer
- This also allows for data to be reformatted or decoded, making additional processing by other systems easier

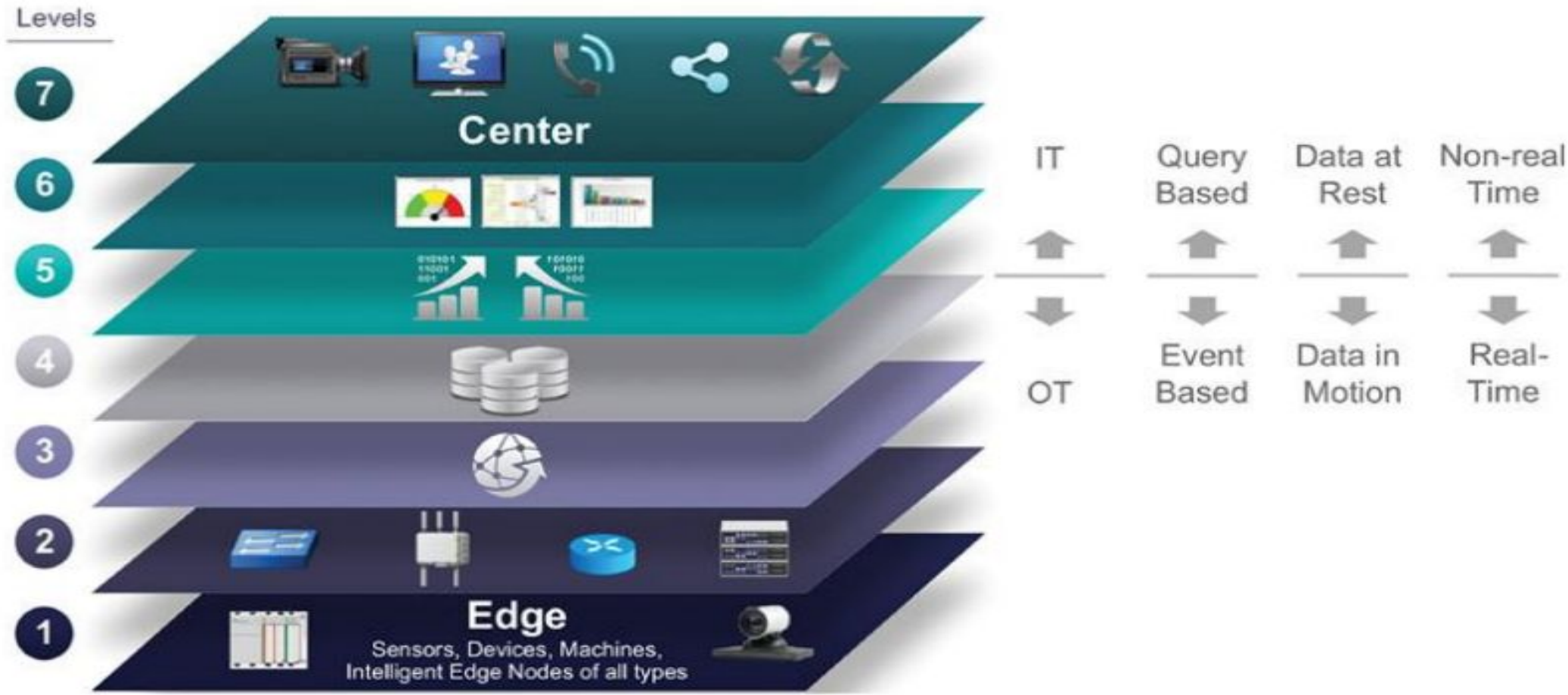

③ Edge (Fog) Computing
(Data Element Analysis and Transformation)
Layer 3 Functions:
• Evaluate and Reformat Data for Processing at Higher Levels
• Filter Data to Reduce Traffic Higher Level Processing
• Assess Data for Alerting, Notification, or Other Actions
Data Ready for Processing at Higher Levels
Data Packets

Upper Layers: Layers 4–7

- The upper layers deal with handling and processing the IoT data generated by the bottom layer

IoT Reference Model Layer	Functions
Layer 4: Data accumulation layer	Captures data and stores it so it is usable by applications when necessary. Converts event-based data to query-based processing.
Layer 5: Data abstraction layer	Reconciles multiple data formats and ensures consistent semantics from various sources. Confirms that the data set is complete and consolidates data into one place or multiple data stores using virtualization.
Layer 6: Applications layer	Interprets data using software applications. Applications may monitor, control, and provide reports based on the analysis of the data.
Layer 7: Collaboration and processes layer	Consumes and shares the application information. Collaborating on and communicating IoT information often requires multiple steps, and it is what makes IoT useful. This layer can change business processes and delivers the benefits of IoT.

IT and OT Responsibilities in the IoT Reference Model



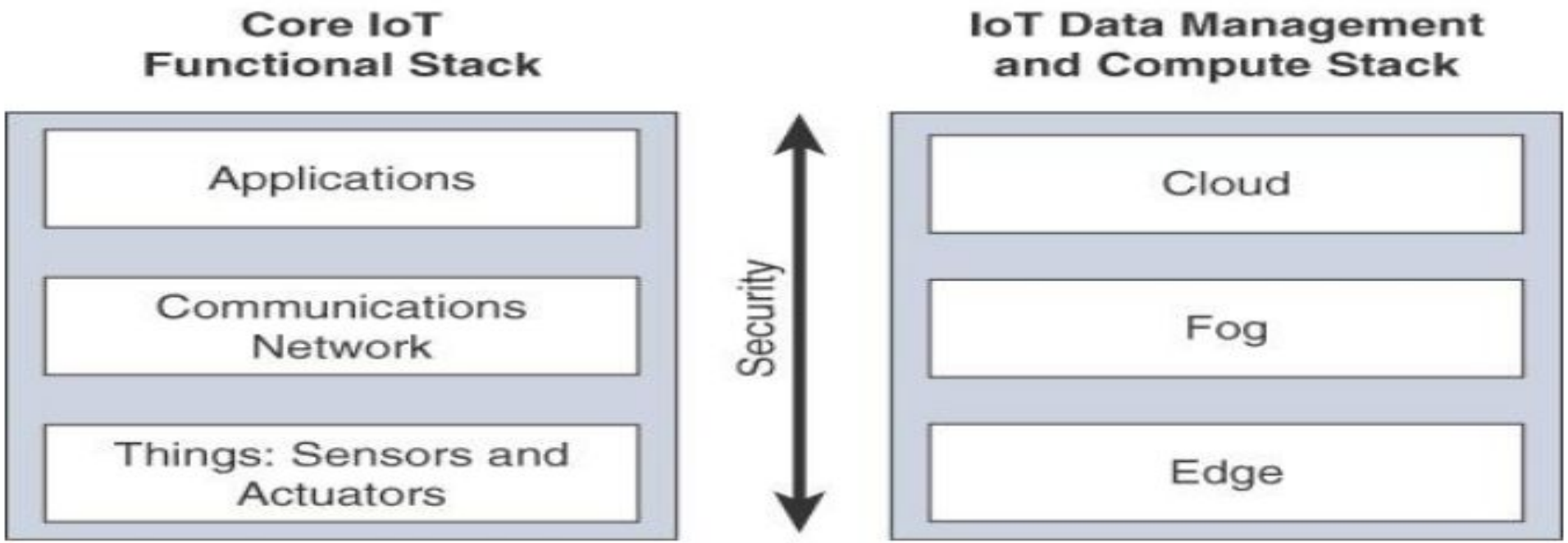
- The network communications layer of the IoT stack itself involves a significant amount of detail and incorporates a vast array of technologies.
- Consider for a moment the heterogeneity of IoT sensors and the many different ways that exist to connect them to a network.
- The network communications layer needs to consolidate these together, offer gateway and backhaul technologies, and ultimately bring the data back to a central location for analysis and processing.

- The network between the gateway and the data center is composed mostly of traditional technologies that experienced IT professionals would quickly recognize.
- These include tunneling and VPN technologies, IP-based quality of service (QoS), conventional Layer 3 routing protocols such as BGP and IP-PIM, and security capabilities such as encryption, access control lists (ACLs), and firewalls

- Unlike with most IT networks, the applications and analytics layer of IoT doesn't necessarily exist only in the data center or in the cloud.
- Due to the unique challenges and requirements of IoT, it is often necessary to deploy applications and data management throughout the architecture in a tiered approach, allowing data collection, analytics, and intelligent controls at multiple points in the IoT system

A SIMPLIFIED IOT ARCHITECTURE

- The framework is presented as two parallel stacks:
- The IoT Data Management and Compute Stack and
- The Core IoT Functional Stack.



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- The three data management layers are
- The edge layer (data management within the sensors themselves),
- The fog layer (data management in the gateways and transit network), and
- The cloud layer (data management in the cloud or central data center).

THE CORE IOT FUNCTIONAL STACK

- IoT networks are built around the concept of "things," or smart objects performing functions and delivering new connected services.
- These objects are "smart" because they use a combination of contextual information and configured goals to perform actions
- These actions can be self-contained (that is, the smart object does not rely on external systems for its actions); however, in most cases, the "thing" interacts with an external system to report information that the smart object collects, to exchange with other objects, or to interact with a management platform

- From an architectural standpoint, several components have to work together for an IoT network to be operational:
- **“Things” layer:**
- At this layer, the physical devices need to fit the constraints of the environment in which they are deployed while still being able to provide the information needed.
- **Communications network layer:**
- When smart objects are not self-contained, they need to communicate with an external system.
- In many cases, this communication uses a wireless technology.
- This layer has four sublayers:

- **Access network sub layer:**
- This is typically made up of wireless technologies such as 802.11ah, 802.15.4g, and LoRa.
- The sensors connected to the access network may also be wired.

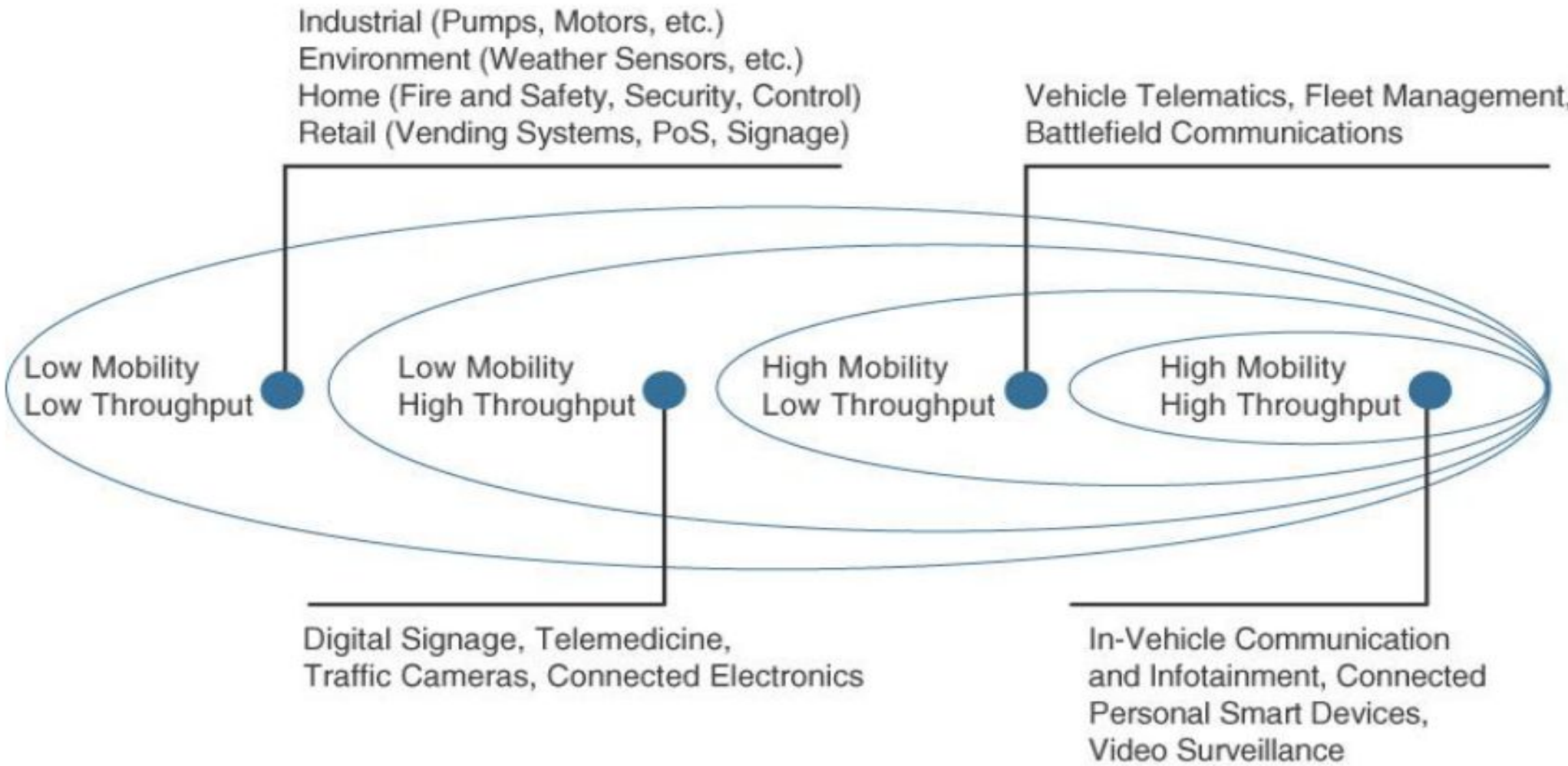
- **Gateways and backhaul network sub layer:**
- The gateway communicates directly with the smart objects.
- The role of the gateway is to forward the collected information through a longer-range medium (called the backhaul) to a headend central station where the information is processed
- **Network transport sub layer:**
- For communication to be successful, network and transport layer protocols such as IP and UDP must be implemented to support the variety of devices to connect and media to use

- **IoT network management sub layer:**
- Additional protocols must be in place to allow the headend applications to exchange data with the sensors.
- Examples include CoAP and MQTT.

- **Application and analytics layer:**
- At the upper layer, an application needs to process the collected data,
- To control the smart objects when necessary,
- To make intelligent decision based on the information collected and, in turn, instruct the “things” or other systems to adapt to the analyzed conditions and change their behaviors or parameters.

Layer 1: Things: Sensors and Actuators Layer

- Battery-powered or power-connected
- Mobile or static
- Low or high reporting frequency
- Simple or rich data
- Report range
- Object density per cell


Industrial (Pumps, Motors, etc.)
Environment (Weather Sensors, etc.)
Home (Fire and Safety, Security, Control)
Retail (Vending Systems, PoS, Signage)
Vehicle Telematics, Fleet Management,
Battlefield Communications
Low Mobility
Low Throughput
Low Mobility
High Throughput
High Mobility
Low Throughput
High Mobility
High Throughput
Digital Signage, Telemedicine,
Traffic Cameras, Connected Electronics
In-Vehicle Communication
and Infotainment, Connected
Personal Smart Devices,
Video Surveillance

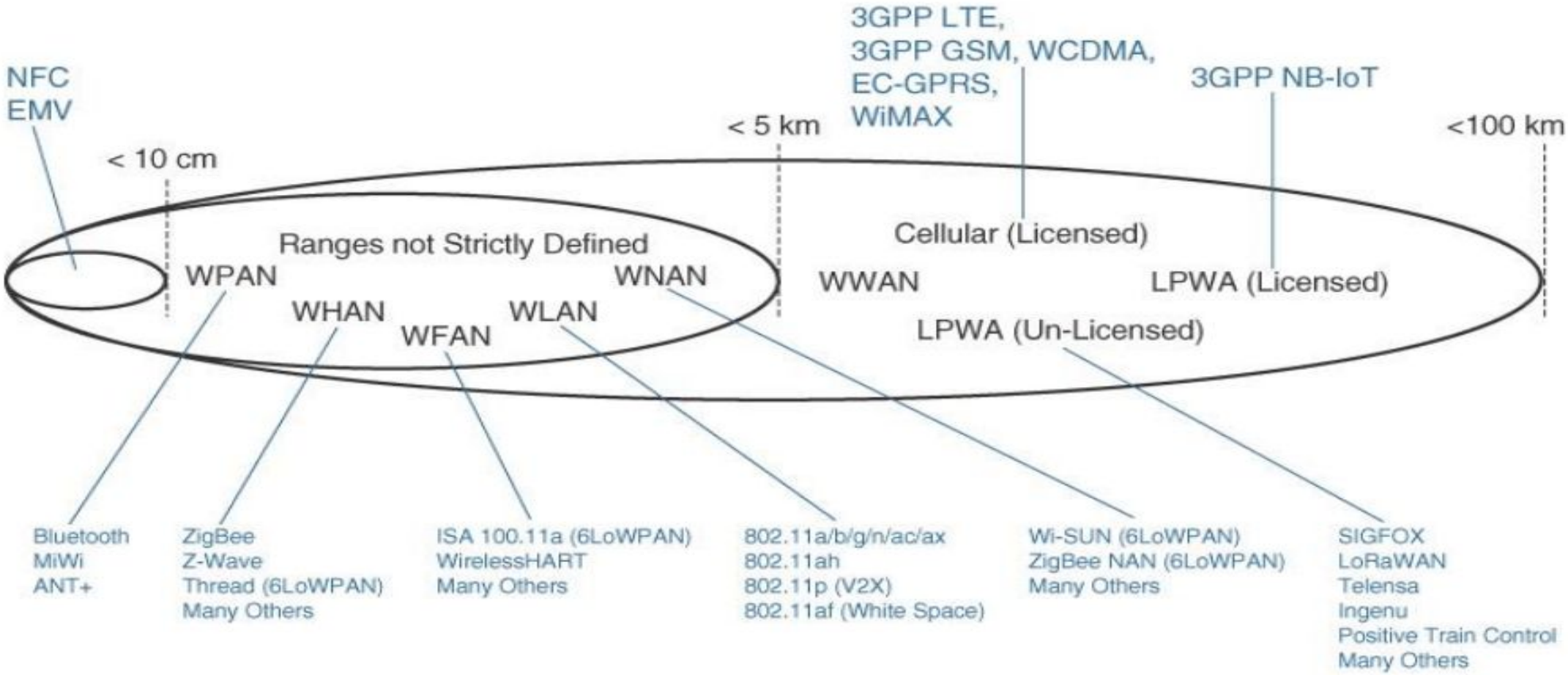
Layer 2: Communications Network Layer

- Once you have determined the influence of the smart object form factor over its transmission capabilities (transmission range, data volume and frequency, sensor density and mobility), you are ready to connect the object and communicate.
- Compute and network assets used in IoT can be very different from those in IT environments
- Major difference is the physical environment in which the devices are deployed

- **Temperature variance**
- Remote external locations, such as those associated with mineral extraction or pipeline equipment can span from the heat of the Arabian Gulf to the cold of the Alaskan North Slope
- Controls near the furnaces of a steel mill obviously require heat tolerance, and controls for cold food storage require the opposite
- In some cases, these controls must handle extreme fluctuations as well
- So, the wide variance in temperature takes a special piece of hardware that is capable of withstanding such harsh environments.

- Humidity fluctuations can impact the long-term success of a system as well.
- Well heads residing in the delta of the Niger River will see very different conditions from those in the middle of the Arabian Desert.
- In some conditions, the systems could be exposed to direct liquid contact such as may be found with outdoor wireless devices or marine condition deployments.

Access Technologies and Distances



WPAN: Wireless Personal Area Network
WHAN: Wireless Home Area Network
WFAN: Wireless Field (or Factory) Area Network
WLAN: Wireless Local Area Network

WNAN: Wireless Neighborhood Area Network
WWAN: Wireless Wide Area Network
LPWA: Low Power Wide Area

Layer 3: Applications and Analytics Layer

- Once connected to a network, your smart objects exchange information with other systems
- **Analytics Versus Control Applications**
- **Analytics application:**
- This type of application collects data from multiple smart objects, processes the collected data, and displays information resulting from the data that was processed
- The important aspect is that the application processes the data to convey a view of the network that cannot be obtained from solely looking at the information displayed by a single smart object

- **Control application:**
- This type of application controls the behavior of the smart object or the behavior of an object related to the smart object
- For example, a pressure sensor may be connected to a pump.
- A control application increases the pump speed when the connected sensor detects a drop in pressure
- Many advanced IoT applications include both analytics and control modules

- **Data Versus Network Analytics**
- Analytics is a general term that describes processing information to make sense of collected data
- **Data analytics:**
- This type of analytics processes the data collected by smart objects and combines it to provide an intelligent view related to the IoT system
- At a very basic level, a dashboard can display an alarm when a weight sensor detects that a shelf is empty in a store.
- In a more complex case, temperature, pressure, wind, humidity, and light levels collected from thousands of sensors may be combined and then processed to determine the likelihood of a storm and its possible path

- **Network analytics:**
- Most IoT systems are built around smart objects connected to the network.
- A loss or degradation in connectivity is likely to affect the efficiency of the system.
- Such a loss can have dramatic effects.
- Loss of connectivity means that data stops being fed to your data analytics platform, and the system stops making intelligent analyses of the IoT system

IoT Data Management and Compute Stack

- The data generated by IoT sensors is one of the single biggest challenges in building an IoT system
- In the case of modern IT networks, the data sourced by a computer or server is typically generated by the client/server communications model, and it serves the needs of the application.
- In sensor networks, the vast majority of data generated is unstructured and of very little use on its own

- In most cases, the processing location is outside the smart object.
- A natural location for this processing activity is the cloud.
- Smart objects need to connect to the cloud, and data processing is centralized.
- One advantage of this model is simplicity.
- Objects just need to connect to a central cloud application.
- That application has visibility over all the IoT nodes and can process all the analytics needed

- This model also has limitations.
- As data volume, the variety of objects connecting to the network, and the need for more efficiency increase, new requirements appear, and those requirements tend to bring the need for data analysis closer to the IoT system
- **Minimizing latency:**
- Milliseconds matter for many types of industrial systems
- Analyzing data close to the device that collected the data can make a difference

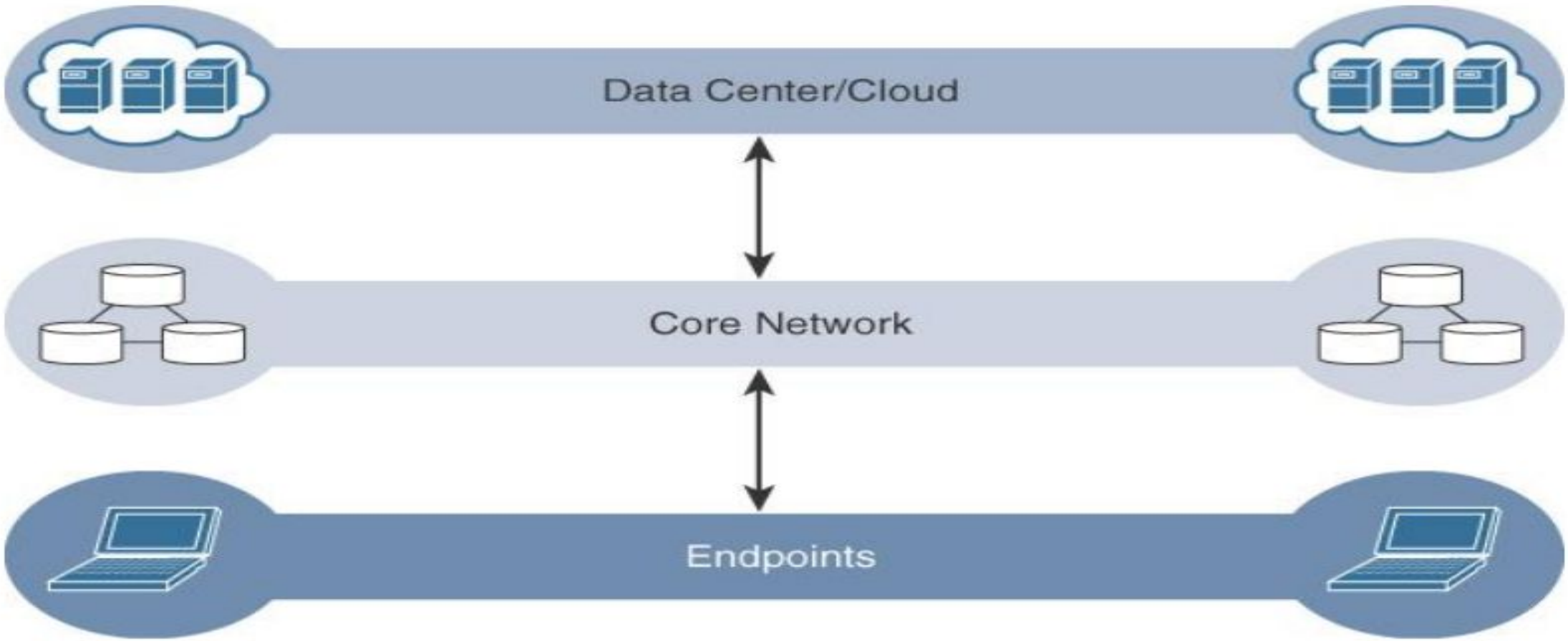
- **Conserving network bandwidth:**
- Offshore oil rigs generate 500 GB of data weekly.
- Commercial jets generate 10 TB for every 30 minutes of flight.
- It is not practical to transport vast amounts of data from thousands or hundreds of thousands of edge devices to the cloud

- **Increasing local efficiency:**
- Collecting and securing data across a wide geographic area with different environmental conditions may not be useful.
- The environmental conditions in one area will trigger a local response independent from the conditions of another site hundreds of miles away.
- Analyzing both areas in the same cloud system may not be necessary for immediate efficiency.

- An important design consideration, therefore, is how to design an IoT network to manage this volume of data in an efficient way such that the data can be quickly analyzed and lead to business benefits.
- The volume of data generated by IoT devices can be so great that it can easily overrun the capabilities of the headend system in the data center or the cloud
- The volume of data also introduces questions about bandwidth management

- Does the network have the capacity to sustain massive volume of traffic?
- Does the application server have the ability to ingest, store, and analyze the vast quantity of data that is coming in?
- “impedance mismatch”

The Traditional IT Cloud Computing Model



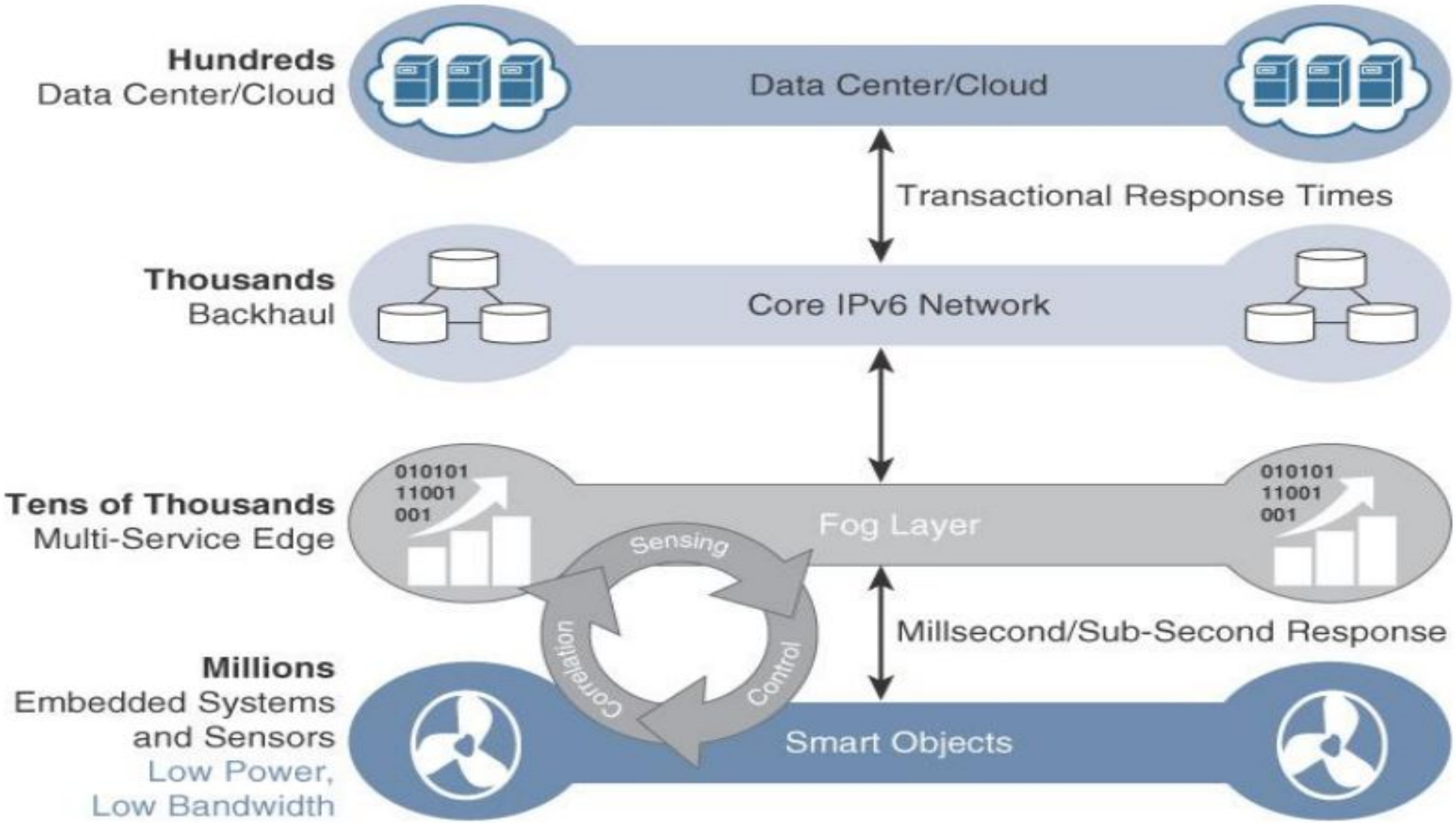
- IoT systems function differently.
- Several data-related problems need to be addressed:
- Bandwidth in last-mile IoT networks is very limited. When dealing with thousands/millions of devices, available bandwidth may be on order of tens of Kbps per device or even less.
- Latency can be very high. Instead of dealing with latency in the milliseconds range, large IoT networks often introduce latency of hundreds to thousands of milliseconds

- Network backhaul from the gateway can be unreliable and often depends on 3G/LTE or even satellite links
- The volume of data transmitted over the backhaul can be high, and much of the data may not really be that interesting
- Big data is getting bigger. The concept of storing and analyzing all sensor data in the cloud is impractical

Fog Computing

- Distribute data management throughout the IoT system, as close to the edge of the IP network as possible.
- The best-known embodiment of edge services in IoT is fog computing.
- Any device with computing, storage, and network connectivity can be a fog node.
- Examples include industrial controllers, switches, routers, embedded servers, and IoT gateways.
- Analyzing IoT data close to where it is collected minimizes latency, offloads gigabytes of network traffic from the core network, and keeps sensitive data inside the local network

The IoT Data Management and Compute Stack with Fog Computing



- Fog services are typically accomplished very close to the edge device, sitting as close to the IoT endpoints as possible.
- One significant advantage of this is that the fog node has contextual awareness of the sensors it is managing because of its geographic proximity to those sensors.
- It can provide contextual analysis of the messages it is receiving and may decide to send back only the relevant information over the backhaul network to the cloud.
- In this way, it is performing distributed analytics such that the volume of data sent upstream is greatly reduced

- In addition, having contextual awareness gives fog nodes the ability to react to events in the IoT network much more quickly than in the traditional IT compute model, which would likely incur greater latency and have slower response times.
- Fog layer -- devices can be monitored, controlled, and analyzed in real time without the need to wait for communication from the central analytics and application servers in the cloud.

- Fog applications are as diverse as the Internet of Things itself.
- What they have in common is data reduction—monitoring or analyzing real-time data from network-connected things and then initiating an action,
- such as locking a door, changing equipment settings, applying the brakes on a train, zooming a video camera, opening a valve in response to a pressure reading, creating a bar chart, or sending an alert to a technician to make a preventive repair

- The defining characteristic of fog computing are as follows:
- **Contextual location awareness and low latency**
- **Geographic distribution**
- **Deployment near IoT endpoints**
- **Wireless communication between the fog and the IoT endpoint**
- **Use for real-time interactions**

Edge Computing

- IoT devices and sensors often have constrained resources, however, as compute capabilities increase.
- Some new classes of IoT endpoints have enough compute capabilities to perform at least low-level analytics and filtering to make basic decisions.

The Hierarchy of Edge, Fog, and Cloud

- It is important to stress that edge or fog computing in no way replaces the cloud
- This model suggests a hierarchical organization of network, compute, and data storage resources.
- At each stage, data is collected, analyzed, and responded to when necessary, according to the capabilities of the resources at each layer.
- As data needs to be sent to the cloud, the latency becomes higher.
- The advantage of this hierarchy is that a response to events from resources close to the end device is fast and can result in immediate benefits, while still having deeper compute resources available in the cloud when necessary.

- It is important to note that the heterogeneity of IoT devices also means a heterogeneity of edge and fog computing resources.
- While cloud resources are expected to be homogenous, it is fair to expect that in many cases both edge and fog resources will use different operating systems, have different CPU and data storage capabilities, and have different energy consumption profiles.

- The most time-sensitive data is analyzed on the edge or fog node closest to the things generating the data.
- Data that can wait seconds or minutes for action is passed along to an aggregation node for analysis and action.
- Data that is less time sensitive is sent to the cloud for historical analysis, big data analytics, and long-term storage